16 Shirley wants to find an estimate for the number of bees in her hive.

On Monday she catches 90 of the bees.
She puts a mark on each bee and returns them to her hive.

On Tuesday she catches 120 of the bees.
She finds that 20 of these bees have been marked.

(a) Work out an estimate for the total number of bees in her hive.

Marked proportion ≈ 20/120 = 1/6.

90/N = 20/120; N = 90 × 120/20 = 540 bees.

3 marks total
Recapture proportion
Set 90/N = 20/120
540 bees

(3)

Shirley assumes that none of the marks had rubbed off between Monday and Tuesday.

(b) If Shirley's assumption is wrong, explain what effect this would have on your answer to part (a).

Rubbed-off marks make 20 too small.

Then 90 × 120/20 is too large: true total is lower.

1 mark
Estimate would be too high

(1)

(Total for Question 16 is 4 marks)

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